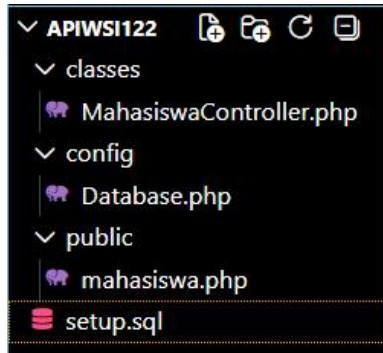


Nama : Fajar Muharram

NIM : E41251215

Gol : D

1. Struktur Folder



2. Setup untuk SQL/Database

```
1 CREATE DATABASE IF NOT EXISTS db_kampus
2 CHARACTER SET utf8mb4
3 COLLATE utf8mb4_unicode_ci;
4
5 USE db_kampus;
6
7 CREATE TABLE IF NOT EXISTS jurusan (
8   id INT AUTO_INCREMENT PRIMARY KEY,
9   kode_jurusan VARCHAR(20) NOT NULL UNIQUE,
10  nama_jurusan VARCHAR(100),
11  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
12  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
13 );
14
15 CREATE TABLE IF NOT EXISTS prodi (
16   id INT AUTO_INCREMENT PRIMARY KEY,
17   kode_prodi VARCHAR(20) NOT NULL UNIQUE,
18   nama_prodi VARCHAR(100),
19   jenjang ENUM('D3','D4','S2') NOT NULL,
20   jurusan INT NOT NULL,
21   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
22   updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
23   INDEX idx_jurusan (jurusan),
24   FOREIGN KEY (jurusan) REFERENCES jurusan(id)
25 );
26
27 CREATE TABLE IF NOT EXISTS mahasiswa (
28   id INT AUTO_INCREMENT PRIMARY KEY,
29   nim VARCHAR(20) NOT NULL UNIQUE,
30   nama VARCHAR(100),
31   prodi INT NOT NULL,
32   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
33   updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
34   INDEX idx_prodi (prodi),
35   FOREIGN KEY (prodi) REFERENCES prodi(id)
36 );
37
38 INSERT INTO jurusan (kode_jurusan, nama_jurusan) VALUES
39 ('ITI', 'Jurusan Teknologi Informasi');
40
41 INSERT INTO prodi (kode_prodi, nama_prodi, jenjang, jurusan) VALUES
42 ('TIF', 'Teknik Informatika', 'D4', 1);
43
44 INSERT INTO mahasiswa (nim, nama, prodi) VALUES
45 ('E41251215', 'Fajar Muharram', 1);
```

3. Membuat Controller Mahasiswa

```
1 <?php
2 class MahasiswaController
3 {
4     private $conn;
5     public function __construct($db)
6     {
7         $this->conn = $db;
8     }
9     public function read_data()
10    {
11        $query = "SELECT * FROM mahasiswa ORDER BY id ASC";
12        $stmt = $this->conn->prepare($query);
13        $stmt->execute();
14        return $stmt->get_result();
15    }
16    public function getDataById($id)
17    {
18        $query = "SELECT * FROM mahasiswa WHERE id = ?";
19        $stmt = $this->conn->prepare($query);
20        $stmt->bind_param("i", $id);
21        $stmt->execute();
22        return $stmt->get_result()->fetch_assoc();
23    }
24    public function store($data)
25    {
26        $query = "INSERT INTO mahasiswa (nim, nama, prodi) VALUES (?, ?, ?)";
27        $stmt = $this->conn->prepare($query);
28        $stmt->bind_param(
29            "sss",
30            $data['nim'],
31            $data['nama'],
32            $data['prodi']
33        );
34        $stmt->execute();
35        return $this->getDataById($this->conn->insert_id);
36    }
37    public function update($id, $data)
38    {
39        if (!$this->getDataById($id)) {
40            return false;
41        }
42
43        $query = "UPDATE mahasiswa SET nim = ?, nama = ?, prodi = ? WHERE id = ?";
44        $stmt = $this->conn->prepare($query);
45        $stmt->bind_param(
46            "sssi",
47            $data['nim'],
48            $data['nama'],
49            $data['prodi'],
50            $id
51        );
52        $stmt->execute();
53        return $this->getDataById($id);
54    }
55    public function destroy($id)
56    {
57        if (!$this->getDataById($id)) {
58            return false;
59        }
60
61        $query = "DELETE FROM mahasiswa WHERE id = ?";
62        $stmt = $this->conn->prepare($query);
63        $stmt->bind_param("i", $id);
64        return $stmt->execute();
65    }
66    public function getAll()
67    {
68        return $this->read_data();
69    }
70 }
71
```

4. Membuat File Mahasiswa.php

```
1 <?php
2 require_once '../config/Database.php';
3 require_once '../classes/MahasiswaController.php';
4
5 $database = new Database();
6 $db       = $database->getConnection();
7 $mahasiswa = new MahasiswaController($db);
8
9 $data = [];
10
11 switch ($_SERVER['REQUEST_METHOD']) {
12
```

A. GET

```
1     case 'GET':
2         $id = $_GET['id'] ?? null;
3
4         if ($id) {
5             $mhs = $mahasiswa->getDataById($id);
6             if ($mhs) {
7                 $data = [
8                     'status' => 'success',
9                     'method' => 'GET',
10                    'message' => 'Data berhasil diambil',
11                    'data'    => $mhs,
12                ];
13            } else {
14                $data = [
15                    'status' => 'error',
16                    'method' => 'GET',
17                    'message' => 'Data tidak ditemukan',
18                ];
19                http_response_code(404);
20            }
21        } else {
22            $mhs = $mahasiswa->getAll();
23            $data = [
24                'status' => 'success',
25                'method' => 'GET',
26                'message' => 'Data berhasil diambil',
27                'count'  => $mhs->num_rows,
28                'data'   => $mhs->fetch_all(MYSQLI_ASSOC),
29            ];
30        }
31        break;
32
33
```

B. POST

```
1 case 'POST':
2     $post = [
3         'nim' => $_POST['nim'] ?? null,
4         'nama' => $_POST['nama'] ?? null,
5         'prodi' => $_POST['prodi'] ?? null,
6     ];
7
8     if ($mhs = $mahasiswa->store($post)) {
9         $data = [
10             'status' => 'success',
11             'method' => 'POST',
12             'message' => 'Data berhasil disimpan',
13             'data' => $mhs,
14         ];
15         http_response_code(201);
16     } else {
17         $data = [
18             'status' => 'error',
19             'method' => 'POST',
20             'message' => 'Gagal menyimpan data',
21         ];
22         http_response_code(400);
23     }
24     break;
25
26
```

C. PUT

```
1 case 'PUT':
2     parse_str(file_get_contents("php://input"), $_PUT);
3     $id = $_PUT['id'] ?? null;
4     $put = [
5         'nim' => $_PUT['nim'] ?? null,
6         'nama' => $_PUT['nama'] ?? null,
7         'prodi' => $_PUT['prodi'] ?? null,
8     ];
9
10    if ($mhs = $mahasiswa->update($id, $put)) {
11        $data = [
12            'status' => 'success',
13            'method' => 'PUT',
14            'message' => 'Data berhasil diupdate',
15            'data' => $mhs,
16        ];
17        http_response_code(200);
18    } else {
19        $data = [
20            'status' => 'error',
21            'method' => 'PUT',
22            'message' => 'Gagal mengupdate data',
23        ];
24        http_response_code(400);
25    }
26    break;
27
```

D. DELETE

```
1  case 'DELETE':
2      $id = $_GET['id'] ?? null;
3
4      if ($mahasiswa->destroy($id)) {
5          $data = [
6              'status' => 'success',
7              'method' => 'DELETE',
8              'message' => 'Data berhasil dihapus',
9              'data' => [],
10         ];
11         http_response_code(200);
12     } else {
13         $data = [
14             'status' => 'error',
15             'method' => 'DELETE',
16             'message' => 'Gagal menghapus data',
17         ];
18         http_response_code(400);
19     }
20     break;
21 default:
22     $data = [
23         'status' => 'error',
24         'message' => 'Metode tidak diizinkan',
25     ];
26     http_response_code(405);
27     break;
28 }
29
30
```

E. mengirim response dalam format JSON ke client

```
1  header('Content-Type: application/json');
2  echo json_encode($data);
3
```

5. Output Localhost

```
localhost/apiws122/public/mahasiswa.php

Pretty-print ☒

{
  "status": "success",
  "method": "GET",
  "message": "Data berhasil diambil",
  "count": 1,
  "data": [
    {
      "id": 1,
      "nim": "E41251215",
      "nama": "Fajar Muharram",
      "prodi": 1,
      "created_at": "2026-05-07 02:28:47",
      "updated_at": "2026-05-07 02:28:47"
    }
  ]
}
```

6. Output Postman

